

Before You Begin:

Install required libraries of R

Two modes are available to the user:

1- All Packages:

This step needs to be performed only once while installing the software and is recommended.

File >> Install R Libraries

2- **Any Single Package:** If any package is missing, user can install that package any time either directly from the default binaries folder for CRAN or Bioconductor packages or directly from the internet

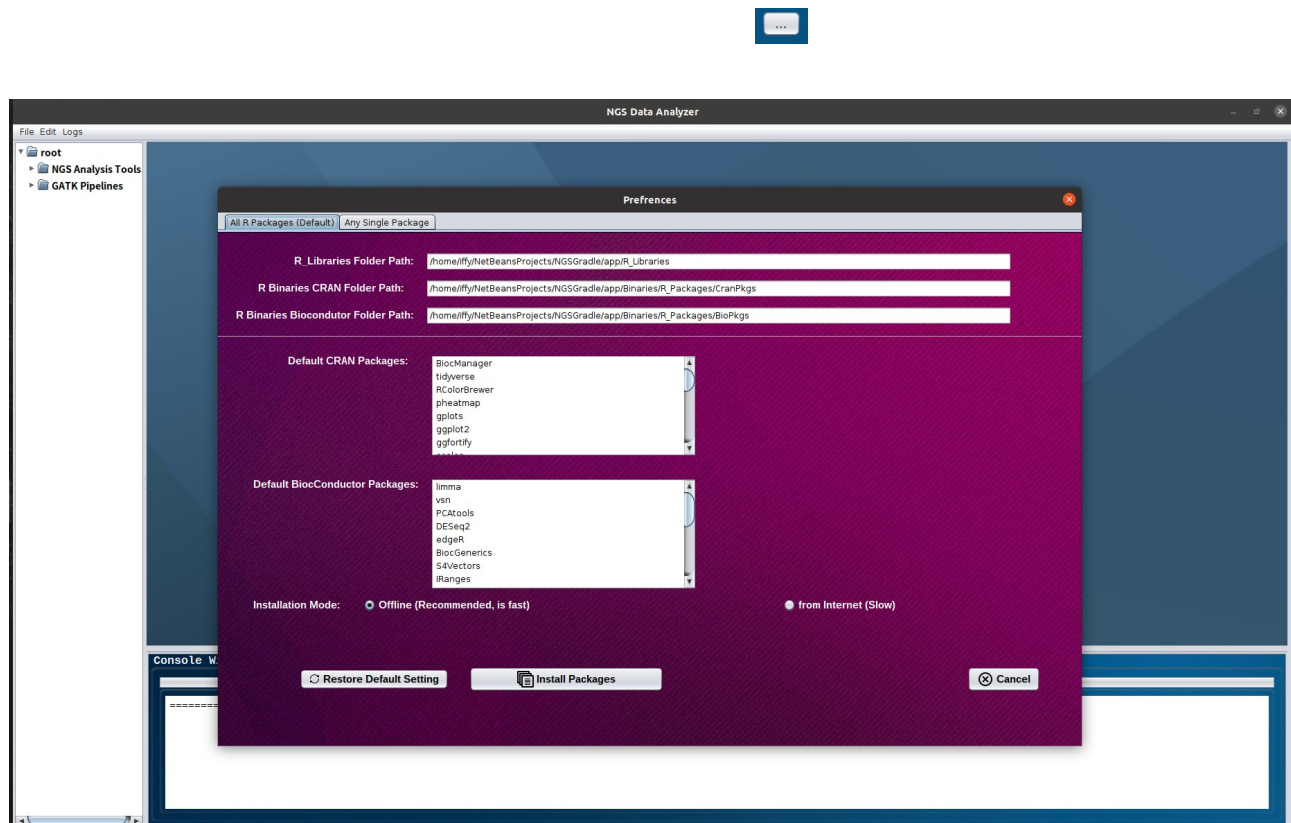


Figure 1: All Packages Installation Dialog (Required only 1st time while installing software)

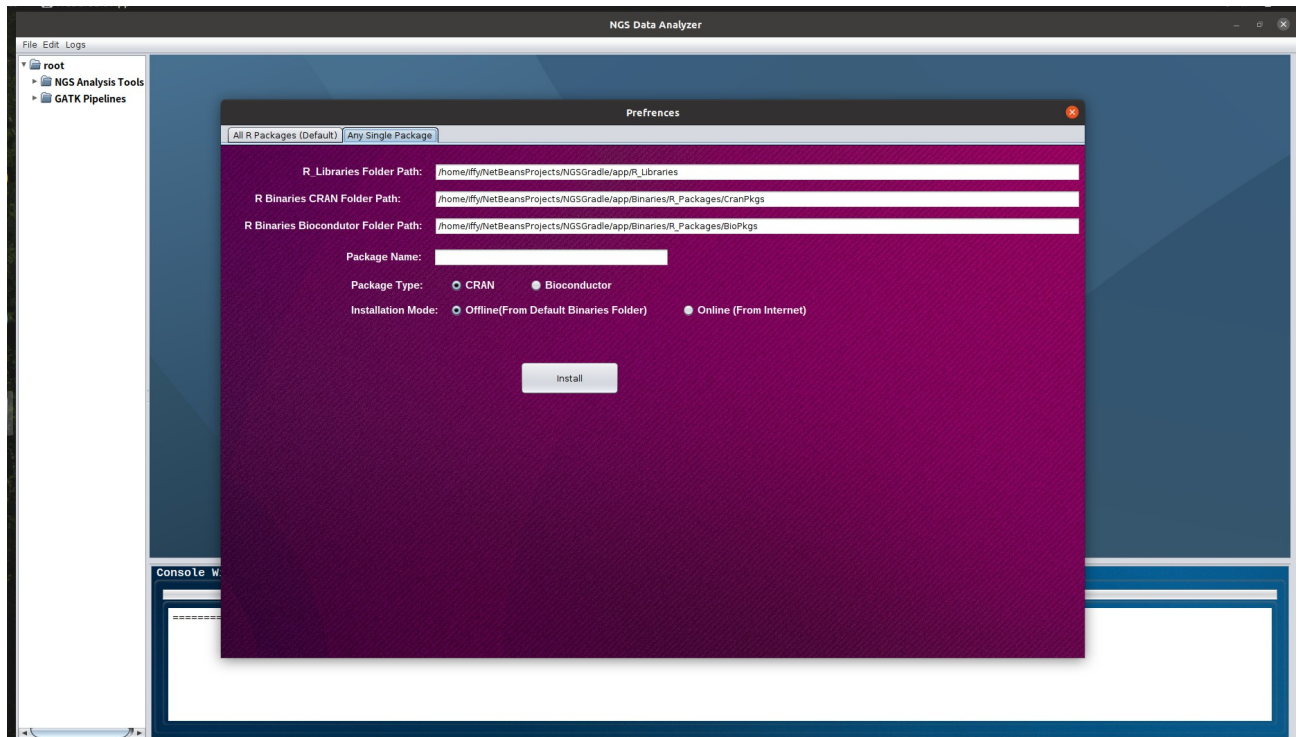


Figure 2: Single Package Installation Dialog (To install Single Packages)

Two installation modes are provided to the user. Offline mode will install libraries from the default pre downloaded binaries and is fast as compared to the other mode. Online mode will download all libraries directly from the internet and install them

Running the Pipelines:

1- RNA Seq Gene Expression Analysis Pipeline:

RNA Seq is a widely used methodology for the study of transcriptome based on NGS that enables to investigate differential gene expression. The correct identification of differentially expressed genes (DEGs) between specific conditions is a key in the understanding phenotypic variation. This pipeline implements a complete graphical user workflow starting from reading fastq reads to calculating featurecounts, performing DEG analysis using DeSeq and EdgeR and finally performing gene set enrichment analysis.

Step 1: Setting up Output Directory and Input files:

User can set up any folder as output directory by clicking and browsing to that folder. All the results generated will be stored in that output folder. Its recommended that folder is empty otherwise files can be overwritten.

NOTE: Path should not contain any spaces.

The screenshot displays the 'RNA SEQ DATA ANALYSIS PIPELINE' window. On the left, a sidebar lists 'root', 'NGS Analysis Tools' (FastQC, STAR, PicardTools, GATK), and 'GATK Pipelines' (RNAseq Variant Discovery Pipeline, RNAseq Pipeline). The main panel has a dark blue background and contains the following elements:

- Header:** 'Enter all the necessary input files. NOTE: Path or Filename should not contain any spaces'
- Set Output Directory:** A text input field with a browse button.
- Select Sample Mapping:** A dropdown menu set to 'Single Sample Per Run (Generate separate bam file for each Sample) (Recommended)'.
- File System:** A dropdown menu set to 'exFAT' with a note: 'NOTE: User can upload input files in *.fq, *.fa, or *.gz zipped format'.
- Input File Selection:**
 - readFilesIn (Single end reads):** A radio button and a text input field labeled 'All sample(s) single end reads' with a browse button.
 - readFilesIn (Paired end reads):** A radio button and two text input fields labeled 'All sample(s) read 1 only' and 'All sample(s) read 2 only', each with a browse button.
- Navigation:** A 'NEXT' button with a right arrow.
- Footer:**
 - 'Already have featureCounts file(s)?'
 - [Click here if want to perform DESEQ & Gene Enrichment Analysis](#)
 - [Click here if want to perform edgeR Analysis](#)

A 'Console Window' is visible at the bottom of the interface.

Figure 3: Setting up output directory and input files path

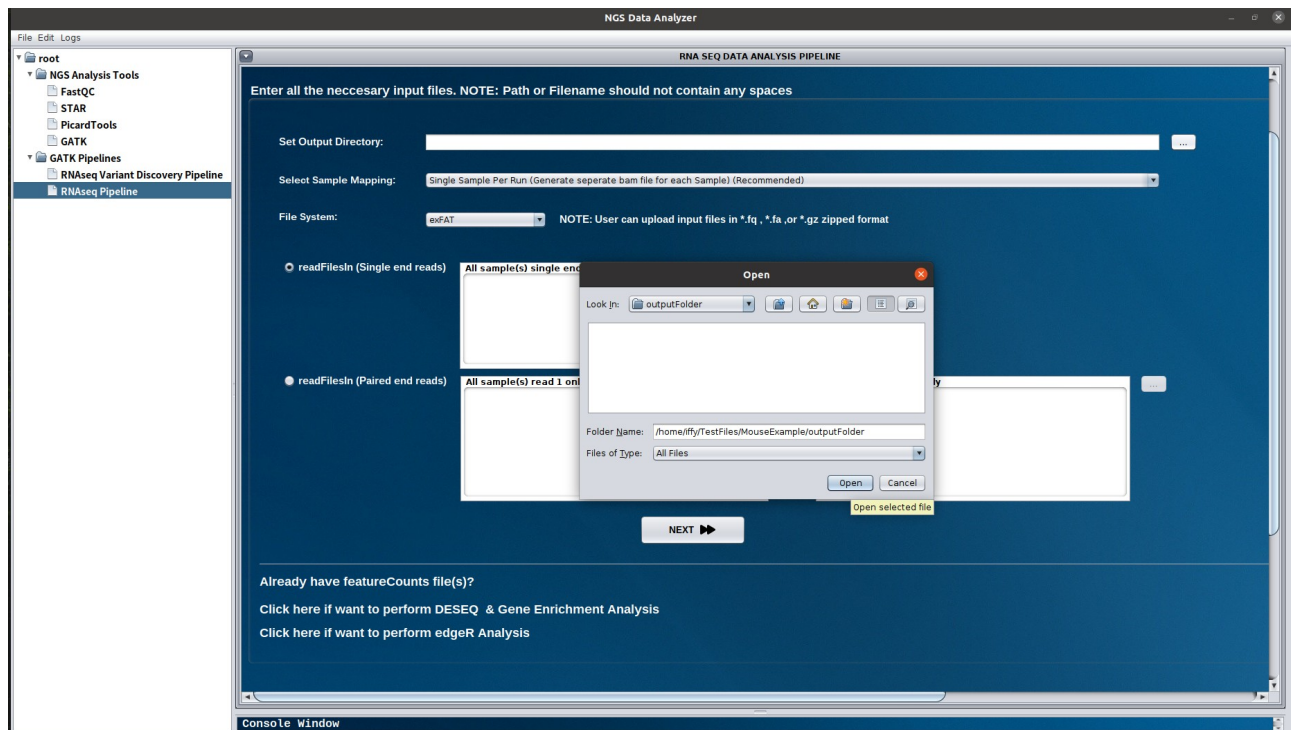
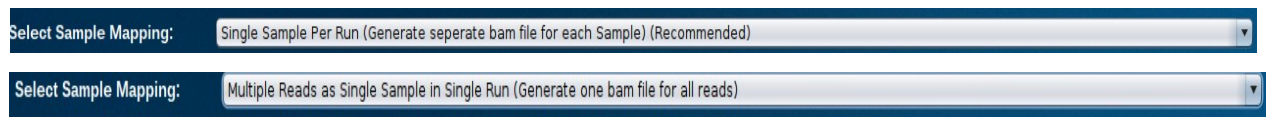


Figure 4: Browsing to output directory

For sample mapping user has two options out of which anyone can be selected. Option “Single Sample Per Run” is recommended:



Next, the user can upload either all single read samples or paired end read samples in any of the supported format (*.fq, *.fa or *.gz)

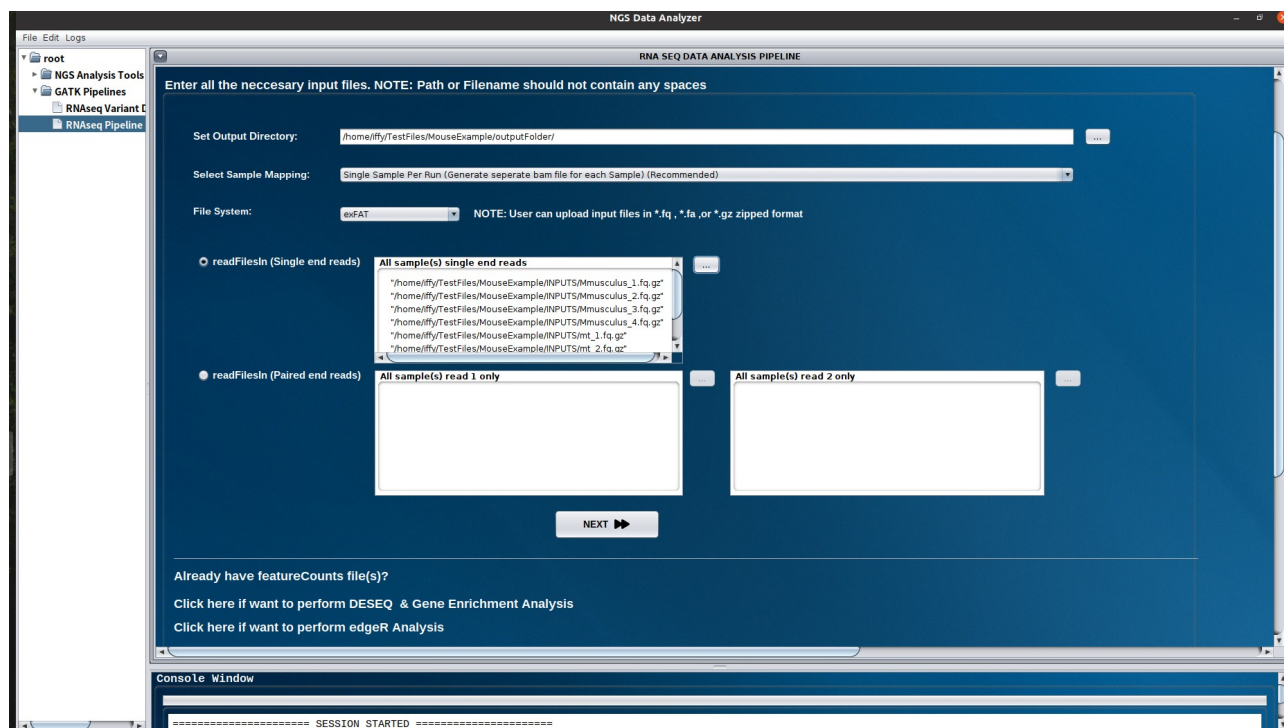


Figure 5: Uploading Read Files (Either Single Reads or Paired End Reads)

Step 2: Quality Check: Read Quality Assessment (Optional)

For quality assessment user can select **FastQC**. Results will be displayed in form of table and HTML reports. HTML reports for each sample can be viewed by clicking drop down menu in HTML results.

The HTML report includes results from multiple modules that were run by FastQC, and provides a quick assessment of the quality of the results labeled as normal (green checkmark), slightly abnormal (orange triangle), and very unusual (red cross) reads. The modules included in the report are as follows:

- **Basic Statistics:** provides introductory compositional statistics such as filename, file type, encoding, total sequences, number of sequences flagged as poor quality, sequence length and %GC for the analyzed read.
- **Per Base Sequence Quality:** displays an overview of the range of quality values across all bases at each position in the FASTQ file. The graph is vertically partitioned into three quality ranges: good (green), reasonable (orange), poor (red).
- **Per Sequence Quality Scores:** displays the quality score distribution over all the sequences, which allows users to see if a subset of the sequences have universally low-quality values.

- **Per Base Sequence Content:** displays the proportion of each DNA base (A, T, C, G) called at a given position in all the sequences.
- **Per Sequence GC content:** displays the GC distribution over all the sequences across the whole length and compares it to a modeled normal distribution of GC content.
- **Per Base N Content:** displays the percentage of base calls at each position for which an N was called. N at a given position indicates the inability to make a normal base call with sufficient confidence.
- **Sequence Length Distribution:** displays the distribution of fragment sizes across all sequences and is highly dependent on the sequencing platform.
- **Sequence Duplication Levels:** counts the degree of duplication for every sequence in a library and creates a plot showing the proportion of sequences with different degrees of duplication (in blue) and de-duplicated sequences (in red).
- **Overrepresented Sequences:** lists all of the sequence which make up more than 0.1% of the total. An overrepresented sequence implies that either it is highly biologically significant, the library is contaminated, or is not that diverse.
- **Adapter Content:** displays the proportion of sequences which have an adapter sequence at a given position. This is informative in deciding whether there is a significant amount of adapter present in the sequences and can be subjected to trimming.
- **Kmer Content:** displays relative K-mer ($k=7$) enrichment over the read length for the top six K-mers. More specifically, it measures the number of each 7-mer at each position in the library and then uses a binomial test to look for significant deviations from an even coverage at all positions and reports the 7-mers with positionally biased enrichment.

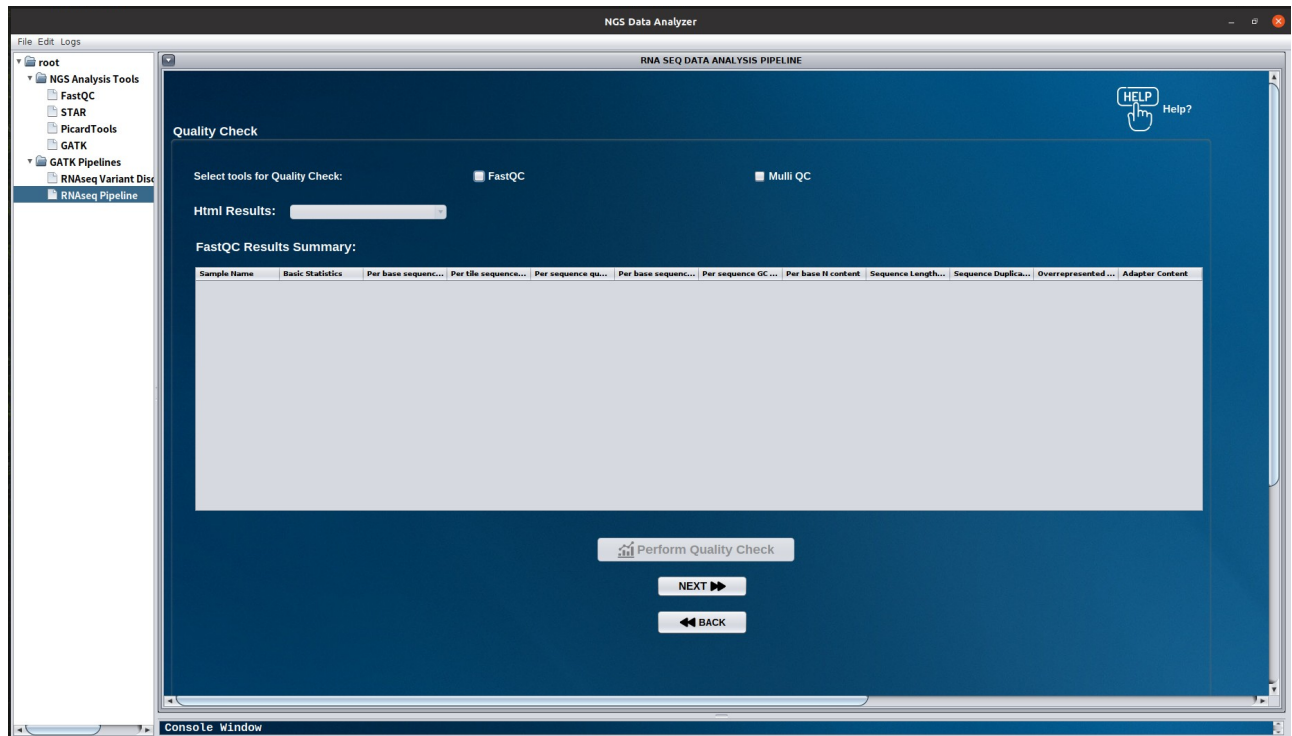


Figure 6: Quality Check through FastQC

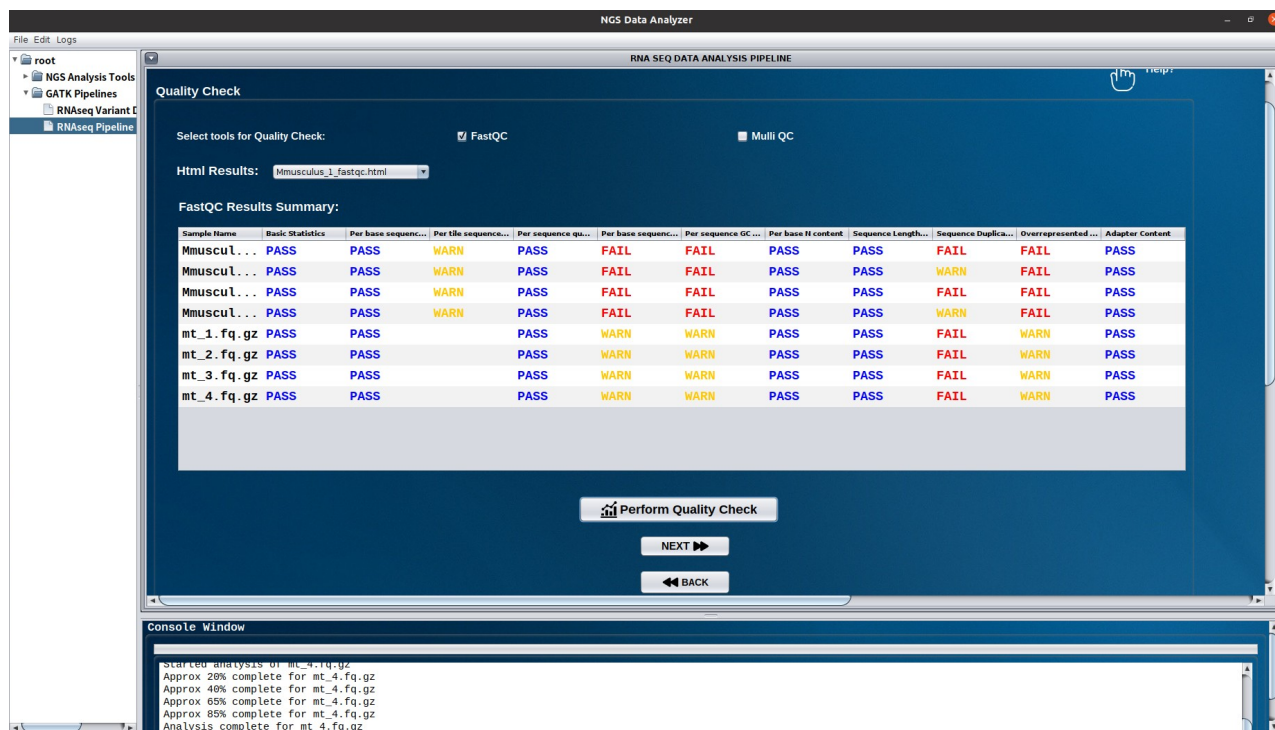


Figure 7: FastQC Tabular Result

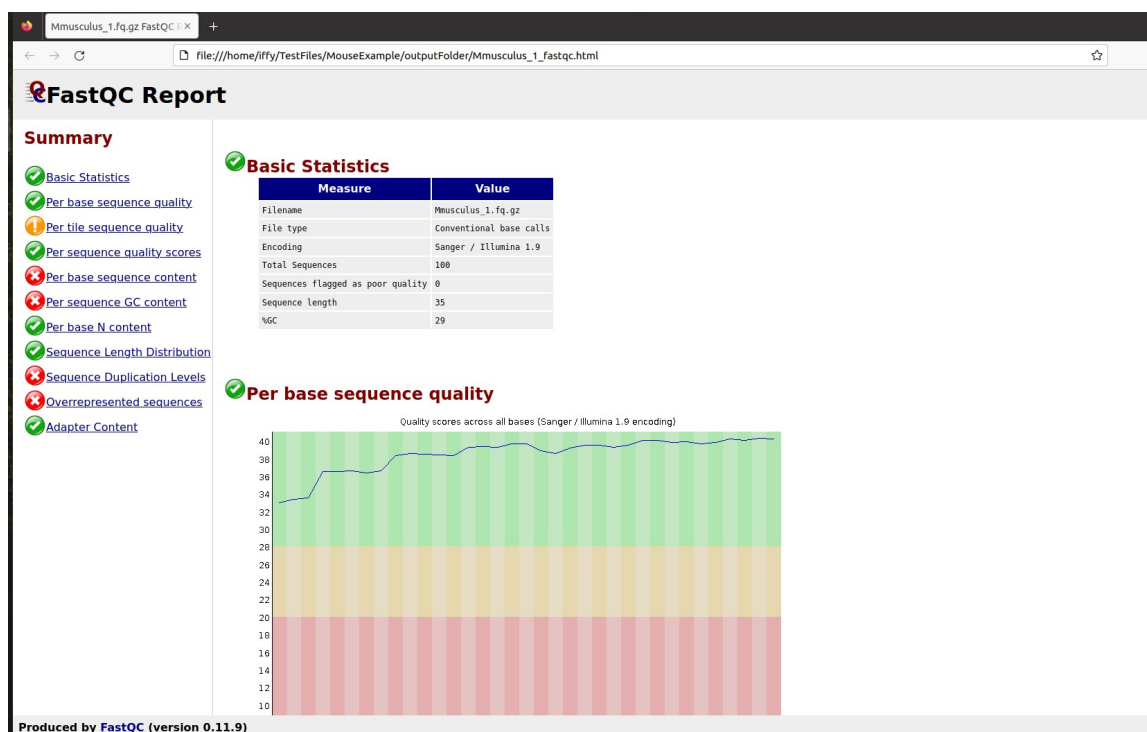
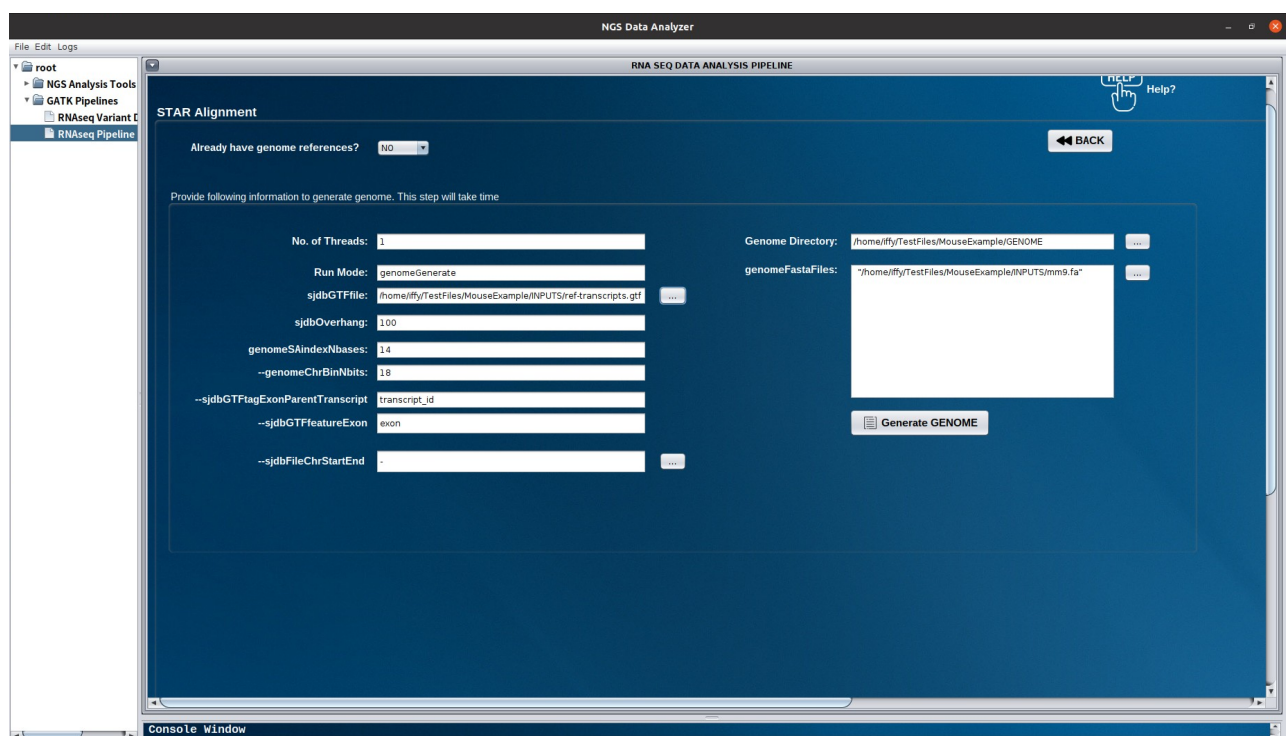


Figure 8: FastQC HTML Report

Step 3: STAR Alignment

If user hasnt already prepared reference genome then 1st user can prepare reference genome by selecting reference genome folder, FASTA reference genome file, GTF file and other STAR parameters accordingly.

This step is only required once.



The screenshot shows the 'STAR Alignment' window of the 'NGS Data Analyzer' application. The window has a dark blue background and a sidebar on the left with a file tree containing 'root', 'NGS Analysis Tools', 'GATK Pipelines', 'RNAseq Variant C', and 'RNAseq Pipeline'. The main area is titled 'STAR Alignment' and contains a form for generating a reference genome. At the top, it asks 'Already have genome references?' with a dropdown set to 'NO' and a 'BACK' button. Below this, it says 'Provide following information to generate genome. This step will take time'. The form includes several input fields: 'No. of Threads' (set to 1), 'Run Mode' (set to 'genomeGenerate'), 'sjdbGTFfile' (set to '/home/itfy/TestFiles/MouseExample/INPUTS/ref-transcripts.gtf'), 'sjdbOverhang' (set to 100), 'genomeSIndexNbases' (set to 14), '--genomeChrBinNbits' (set to 18), '--sjdbGTFtagExonParentTranscript' (set to 'transcript_id'), '--sjdbGTFfeatureExon' (set to 'exon'), and '--sjdbFileChrStartEnd' (set to 5). There are also buttons for 'Genome Directory' and 'genomeFastaFiles' pointing to specific paths. A 'Generate GENOME' button is at the bottom right. A 'Console Window' is visible at the bottom of the application.

Figure 9: Indexing Reference Genome (This step will take time)

If user has already prepared reference genome then user can directly start alignment of raw reads with reference genome with appropriate STAR parameters.

Also user can chose either to create separate featureCounts file for each sample or to generate a combined featureCounts file for all samples.

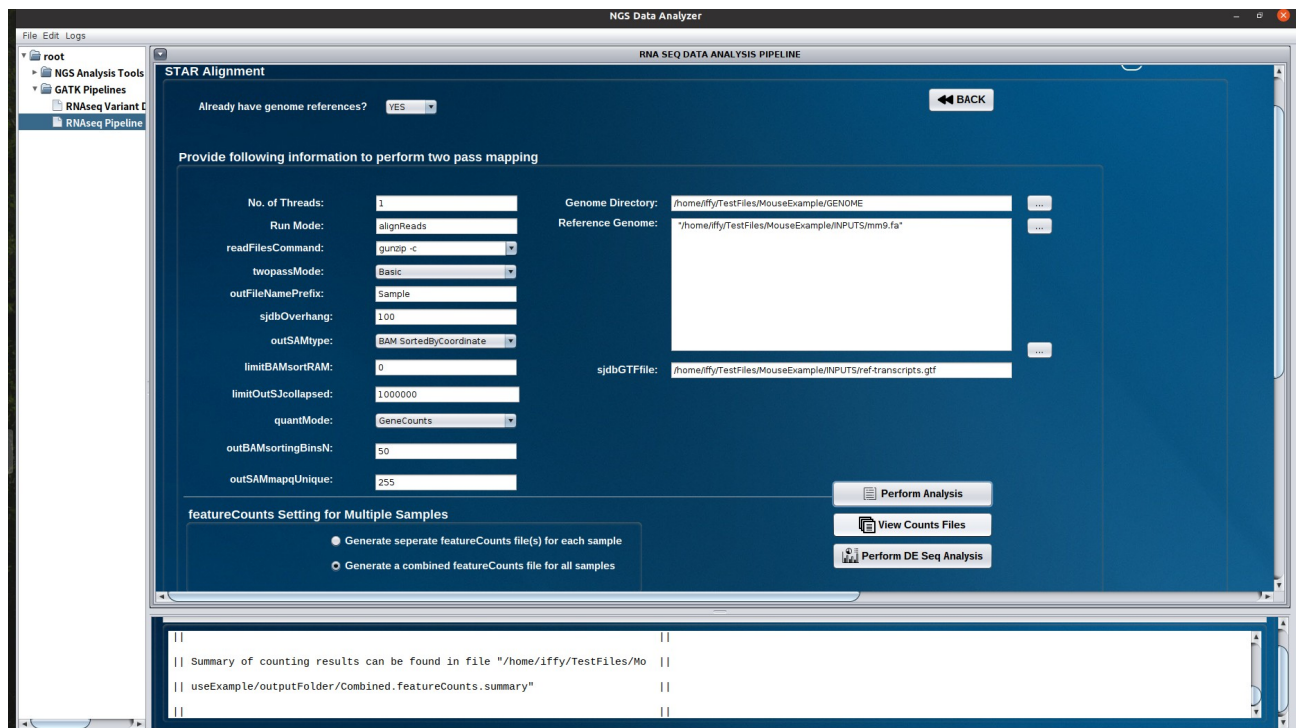
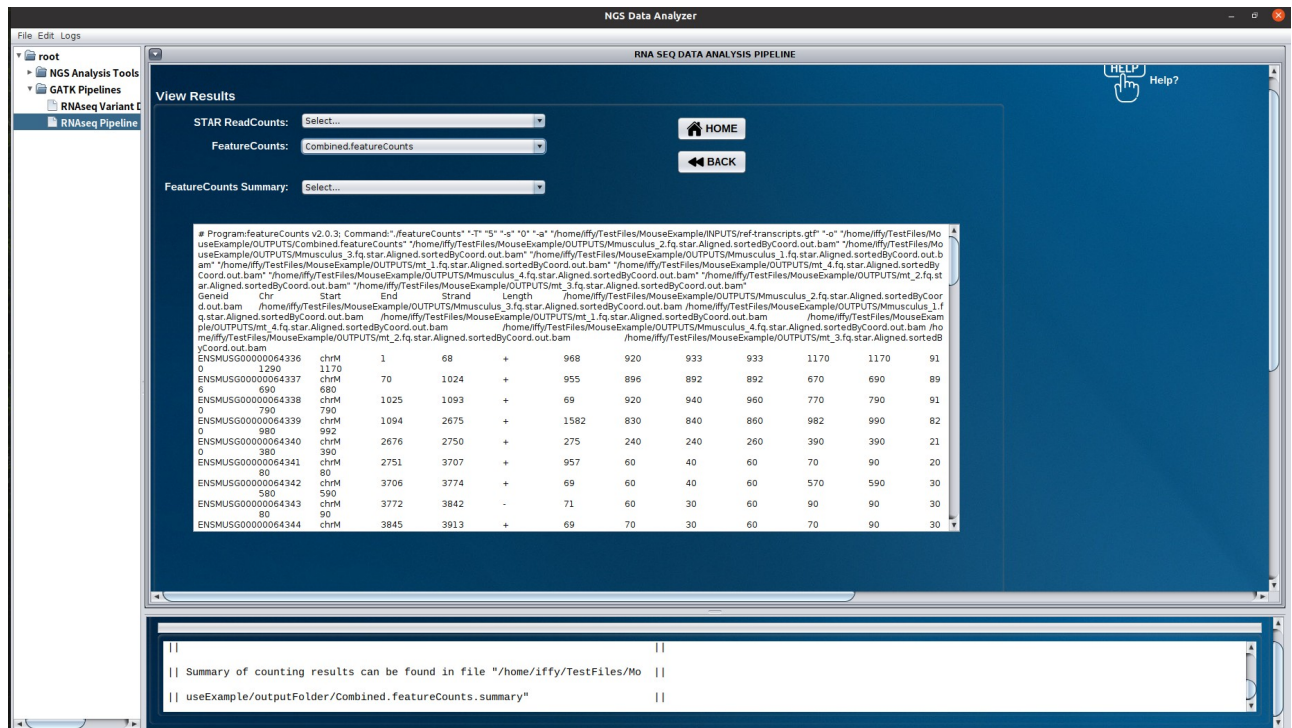


Figure 10: Two Pass mapping and Generation of Count Files

View counts File:

To view the resulting count files user can check the output directory or can also view through “View Count Files” option



Step 5: DE Seq Analysis

Once featureCounts files are created user can start with DESEQ2 analysis. User can select mode of analysis for single or multiple factors. **Factors** are categorical variables or classifiers associated with samples in the experiment.

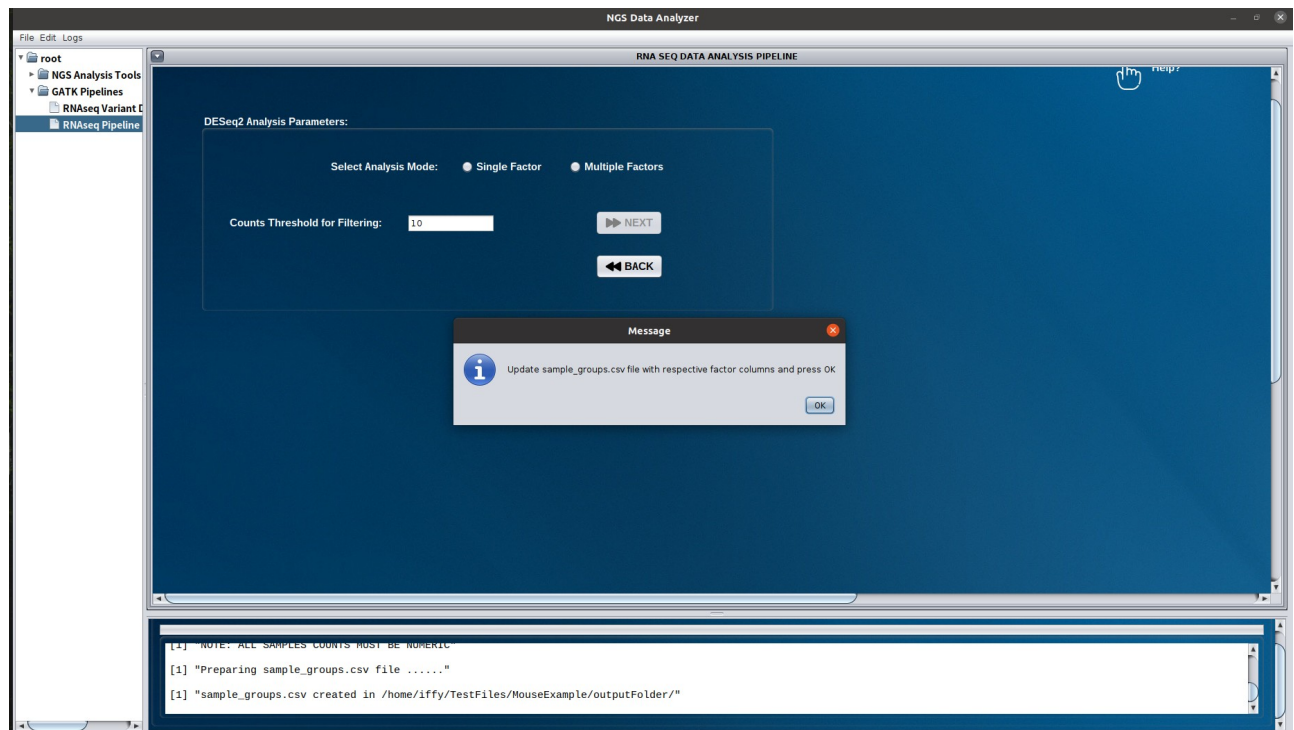


Figure 12: Perform DESeq Analysis (sample_groups.csv will be generated in output directory)

A file named **sample_groups.csv** will be generated in the output directory containing all the samples names. User can incorporate factors information as columns right next to sample names in the csv file. Once **sample_groups.csv** file is modified then user can move on to next step

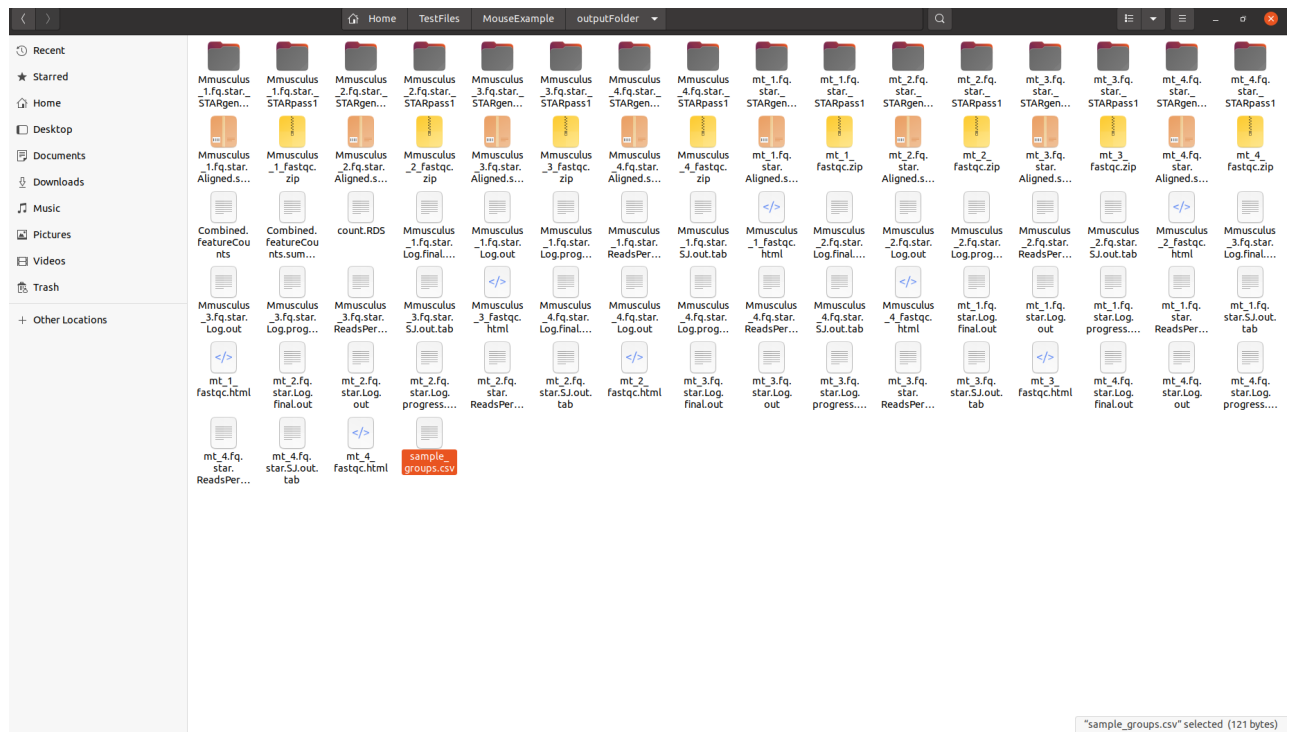


Figure 13: Output Directory Example

sample_groups.csv - LibreOffice Calc

SampleName																		
1	SampleName																	
2	Mmusculus_2.fq																	
3	Mmusculus_3.fq																	
4	Mmusculus_1.fq																	
5	mt_1.fq																	
6	mt_4.fq																	
7	Mmusculus_4.fq																	

sample_groups.csv - LibreOffice Calc

SampleName	Type	Gender																
1	SampleName																	
2	Mmusculus_2.fq	treated	Male															
3	Mmusculus_3.fq	treated	Female															
4	Mmusculus_1.fq	treated	Male															
5	mt_1.fq	untreated	Male															
6	mt_4.fq	untreated	Female															
7	Mmusculus_4.fq	treated	Female															
8	mt_2.fq	untreated	Male															
9	mt_3.fq	untreated	Female															

Once user has added the factors information in sample_groups.csv then user can select appropriate options according to factors.

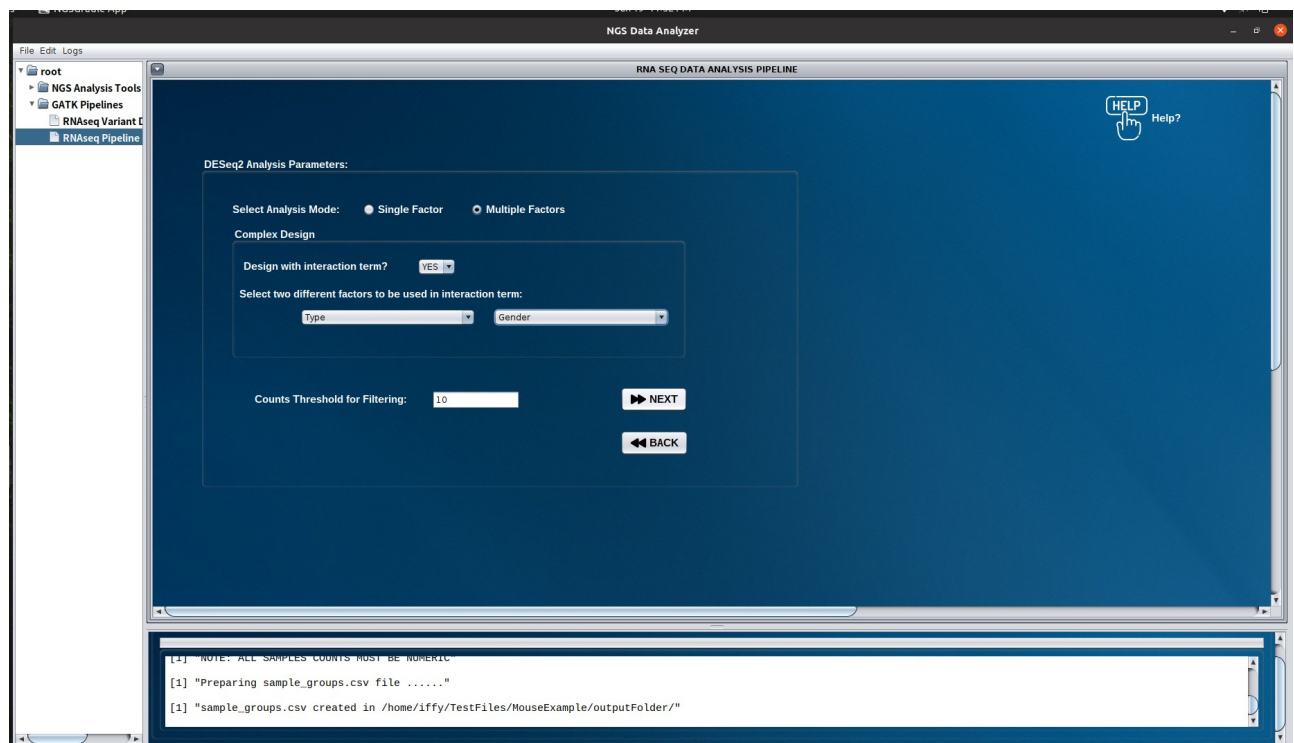


Figure 14: DESeq Analysis (Select factors and start the analysis)

Depending upon the information provided in sample_groups.csv user can chose between single (Simple design) or multiple factor (Complex Design). Note that multiple factor right now includes two factors. Filtering genes with less count prior to analysis is important in order to decrease the impact on false discovery rates so user can set a counts threshold value for filtering. User can click next to perform differential expression analysis.

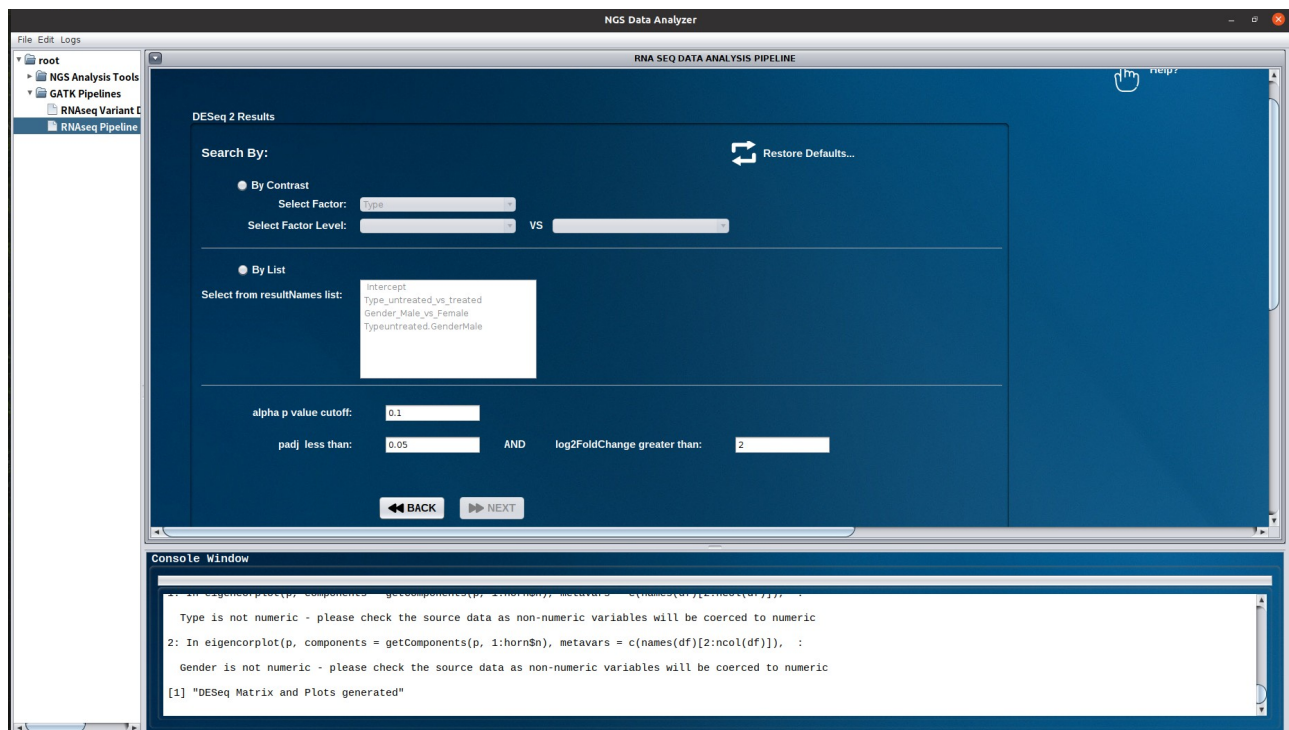


Figure 15: Analyze DESeq Results

Once the analysis is complete now user can easily explore the results by checking folder named **DeSeqResults** in output directory. All the graphs for DESeq analysis will be stored in that folder. Two files named as DESeq2_Results.txt and DESeq_results.csv will also be generated containing all necessary information about the analysis. User can further filter out significant genes on the bases of **padj** and **log2foldchange** values as per results. Make sure to provide a range according to results obtained otherwise error will be generated. Its necessary to obtain significant genes in result to perform next gene enrichment analysis.

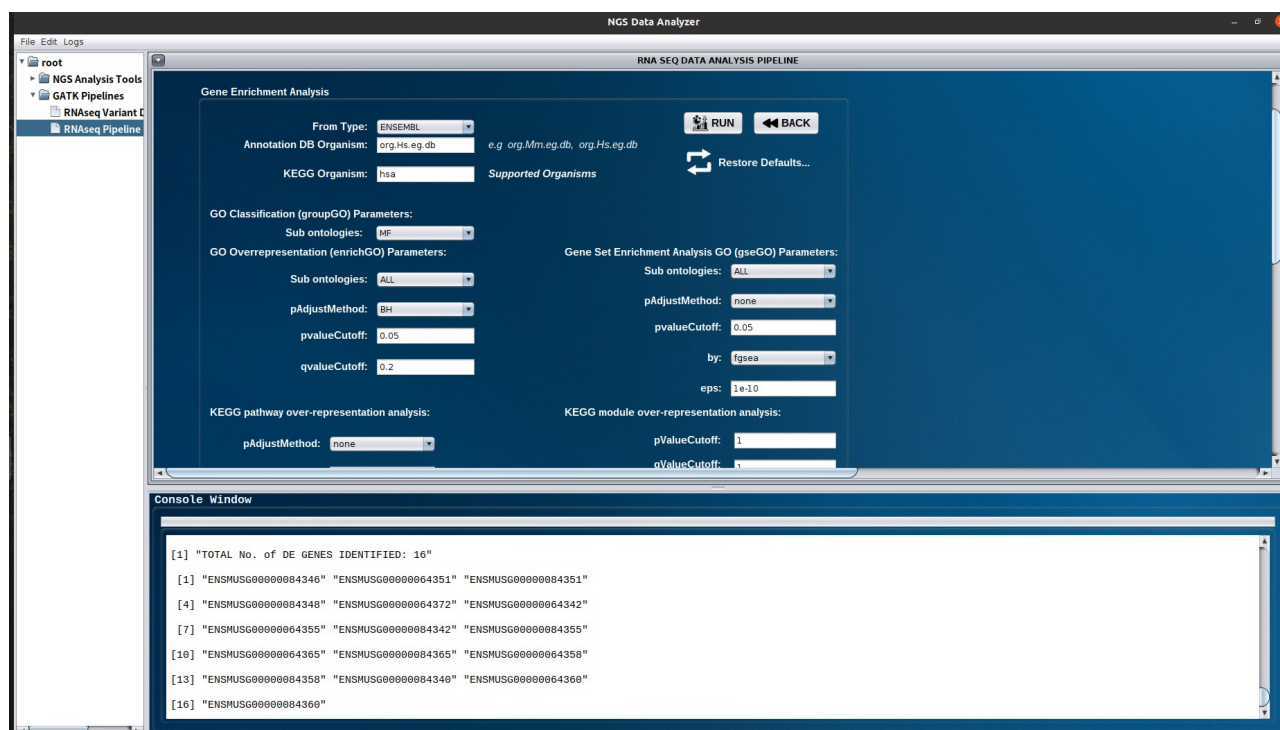
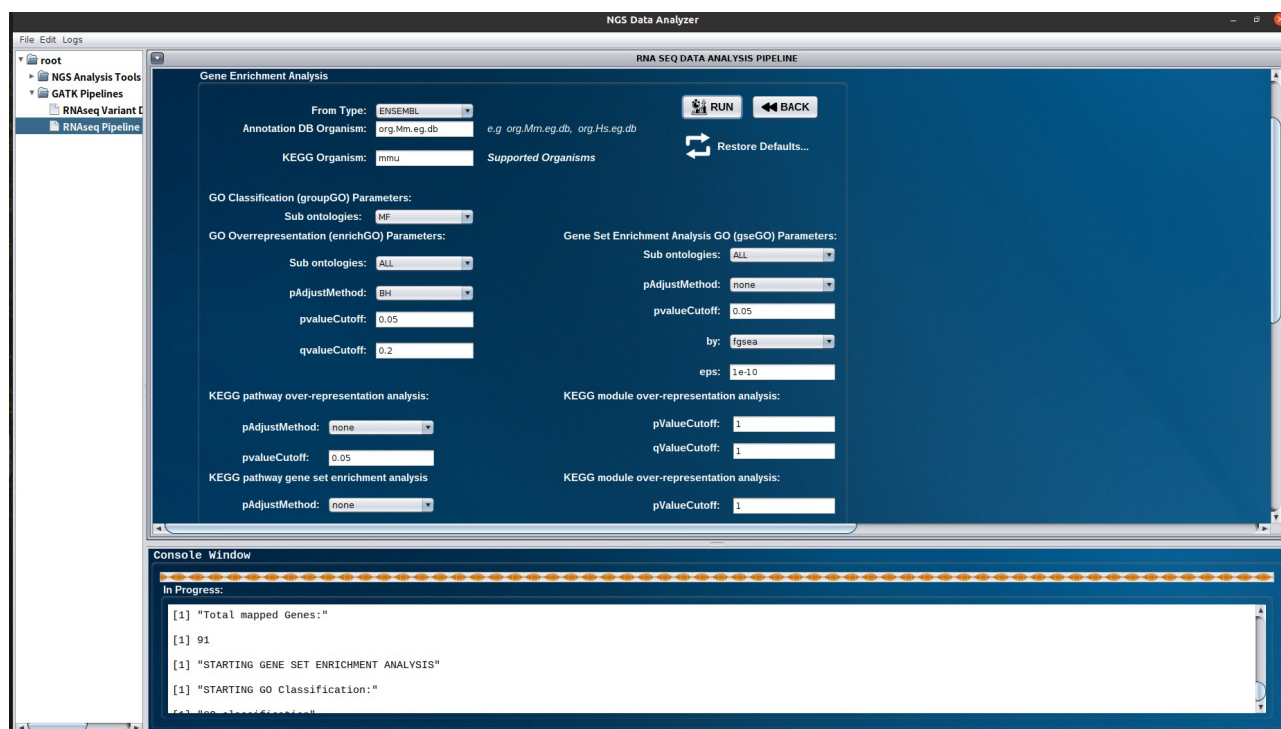


Figure 16: Gene Expression Analysis (After finalizing significantly expressed genes according to users criteria)

Once differentially expressed genes are identified, user can proceed to gene enrichment analysis interface.



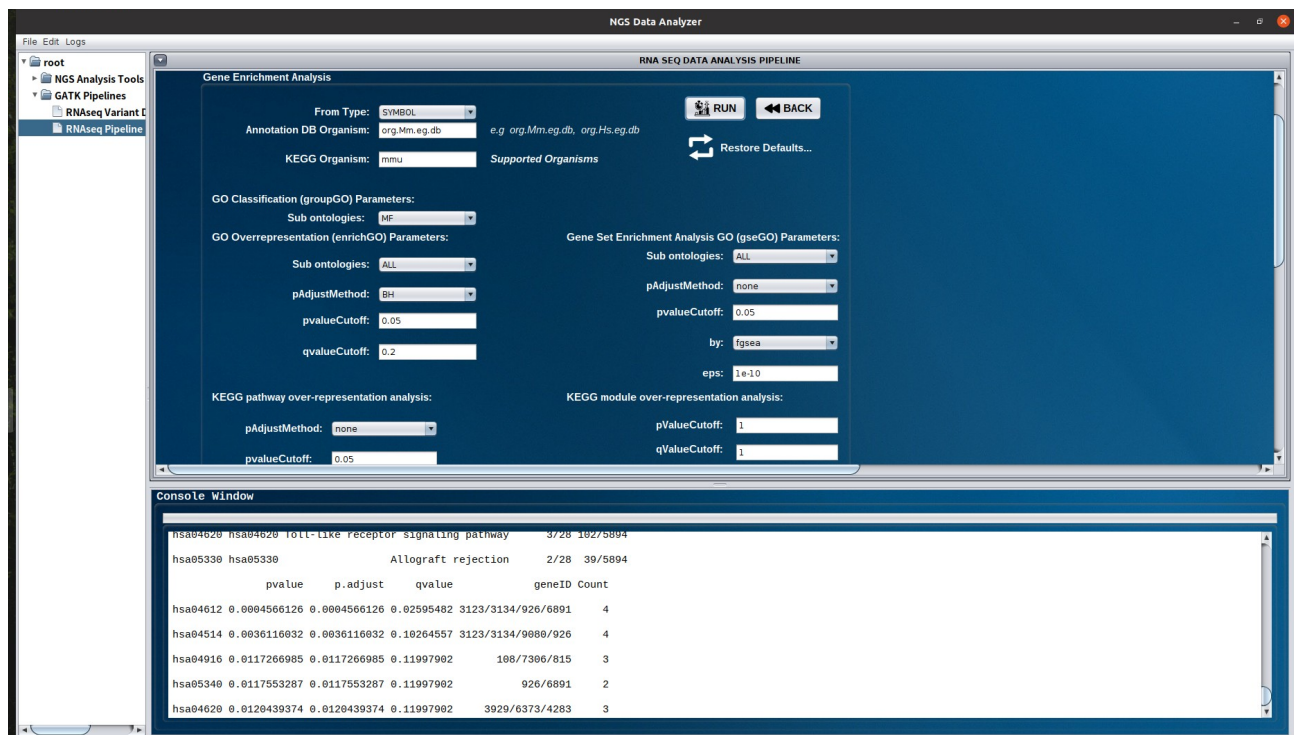
Gene Set Enrichment Analysis (GSEA) is a computational method used to analyze high-throughput gene expression data, to identify gene sets or biological pathways that are significantly associated with a particular phenotype or experimental condition.

From Type option allows user to chose the format type of genes that is Ensemble, Symbol, Uniprot, EntrezID.

Annotation DB organism option requires a valid database provided by bioconductor AnnotationData packages (<https://bioconductor.org/packages/3.17/data/annotation/>).

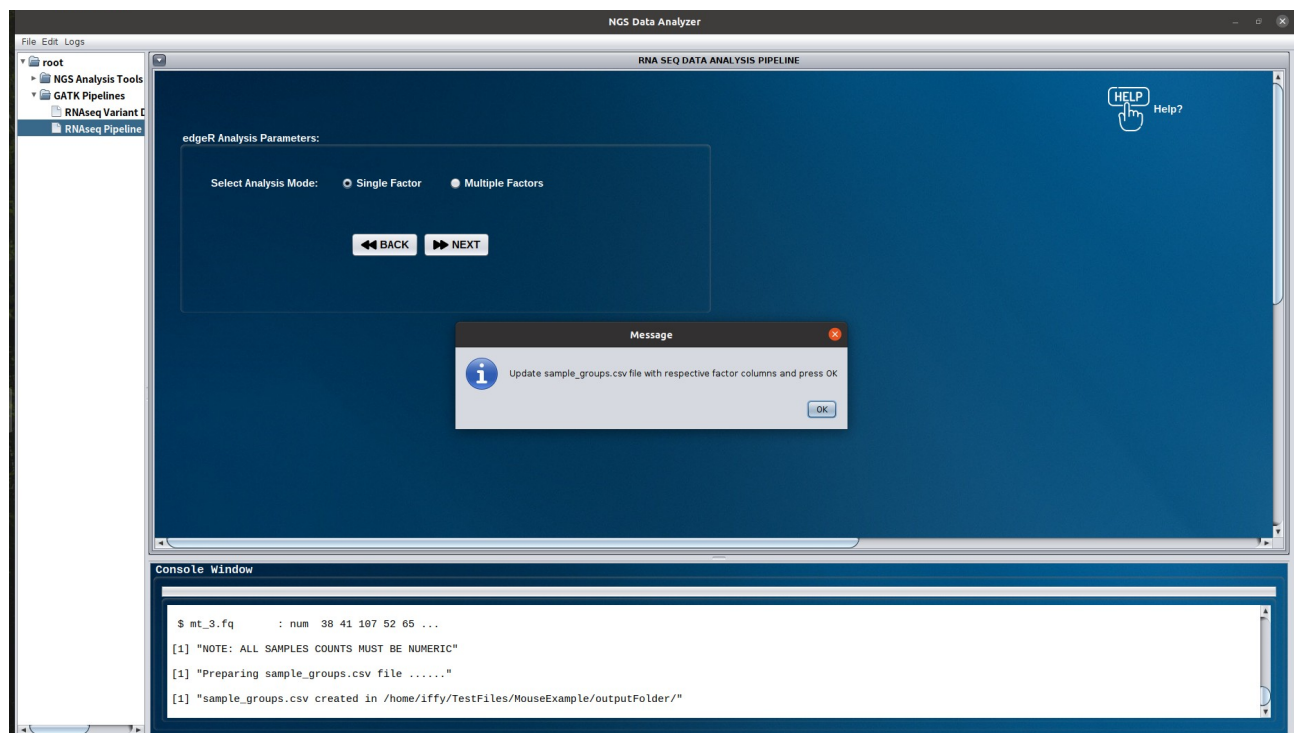
KEGG Organism requires a valid organism identifier from https://www.genome.jp/kegg/catalog/org_list.html

User can set rest of the parameters accordingly.

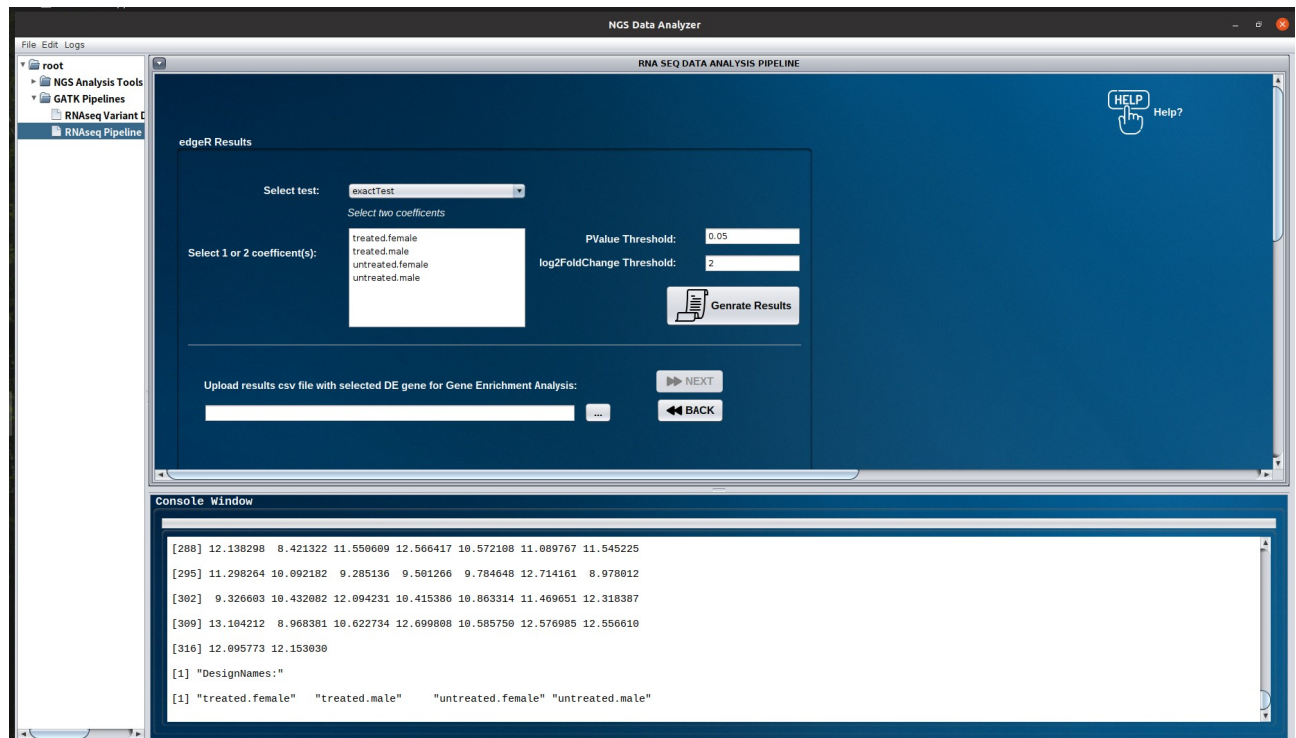


EdgeR Analysis:

Initial Interface:



Modify the sample_groups.csv file same as explained above and select factors. Once factors are selected click next to start edgeR analysis.



Once analysis is complete user can perform different test on the results including exactTest, glmQLFTest, glmTreat and glmLRT.

All results will be saved in **EdgeRResults** folder inside output directory.

Once the user has uploaded results csv for example exactTest.csv having significant genes information, he can perform gene enrichment analysis in the same manner as discussed above